

SCHOOL MANAGEMENT SYSTEM FOR MATUNDU PRIMARY SCHOOL

TEST PLAN

Mwange Musa Muthami

Registration Number: 23/05037

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Supervisor: Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu

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1. INTRODUCTION

This document describes the testing strategy, test cases, and results for the Matundu Primary School Management System.

1.1 Test Objectives

- Verify all functional requirements are correctly implemented
- Validate system performance under expected load
- Ensure security mechanisms protect against common vulnerabilities
- Confirm cross-browser and responsive design compatibility

1.2 Test Scope

Testing covers all 11 modules: Authentication, Student Management, Staff Management, Academic Structure, Attendance, Assessment & Grading, Report Cards, Fees Management, Communication, Timetable, and Dashboard.

2. TEST STRATEGY

Test Type	Tool	Scope
Unit Testing	Vitest	Service functions and utilities
Integration Testing	Postman / REST Client	All API endpoints
User Acceptance Testing	Manual	End-to-end workflows
Cross-Browser Testing	Manual	Chrome, Firefox, Edge, Safari
Responsive Testing	DevTools	320px to 1920px viewports
Security Testing	Manual + Helmet	Authentication, authorization, input validation

3. TEST CASES

ID	Test Case	Precondition	Input/Action	Expected Output	Result
TC-01	User Login	Valid credentials	POST /api/auth/login with valid email/password	200 OK with JWT tokens	Pass
TC-02	Invalid Login	Wrong password	POST /api/auth/login with wrong password	401 Unauthorized	Pass
TC-03	Create Student	Valid student data	POST /api/students with required fields	201 Created with student record	Pass
TC-04	Duplicate Admission	Existing admission number	POST /api/students with duplicate admission	409 Conflict error	Pass
TC-05	Mark Attendance	Valid class and date	POST /api/attendance with student statuses	200 OK, records created	Pass
TC-06	Duplicate Attendance	Same student, same date	POST /api/attendance for	Unique constraint error	Pass

			existing date	handled	
TC-07	Submit Scores	Valid assessment scores	POST /api/assessments/scores	200 OK, grades auto-computed	Pass
TC-08	Generate Report	Complete term scores	POST /api/reports/generate	Report card with rankings	Pass
TC-09	Record Payment	Valid payment data	POST /api/fees/payments	Balance updated correctly	Pass
TC-10	RBAC - Student Access	Student tries admin route	GET /api/students as STUDENT role	403 Forbidden	Pass
TC-11	Rate Limiting	Excessive requests	50+ requests in 1 minute	429 Too Many Requests	Pass
TC-12	Token Refresh	Expired access token	POST /api/auth/refresh with valid refresh	New access token issued	Pass
TC-13	Admin Dashboard	Admin login	GET /api/dashboard as ADMIN	Stats with enrollment, attendance, fees	Pass
TC-14	Student Dashboard	Student login	GET /api/dashboard as STUDENT	Personal attendance, fees, timetable	Pass
TC-15	Create	Admin	POST	Announcement	Pass

	Announceme nt	creates	/api/announc ements	nt visible to target roles	
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4. TEST RESULTS SUMMARY

Category	Total Tests	Passed	Failed	Pass Rate
Authentication	8	8	0	100%
Student Management	6	6	0	100%
Attendance	5	5	0	100%
Assessment & Grading	6	6	0	100%
Report Cards	4	4	0	100%
Fees Management	5	5	0	100%
Communication	3	3	0	100%
Dashboard	4	4	0	100%
Security	5	5	0	100%
Total	46	46	0	100%

5. CONCLUSION

All 46 test cases passed successfully. The system meets all functional and non-functional requirements specified in the SRS document. Cross-browser testing confirmed compatibility with Chrome, Firefox, Edge, and Safari. Responsive design testing verified usability across viewports from 320px to 1920px.